

Prj 1: Multi-Robot System in Cooperative Object Transportation Using Deep Reinforcement Learning

Computation Intelligence and its Applications in Mechatronics
Amirkabir University of Technology



In this project, you are supposed to design, implement, and train a multi-robot system to collaboratively transport a long rod-shaped object from one location to another using deep reinforcement learning.

Project Structure

1. Task 1: Coordination with Fixed Attachment — The robots start with the rod already fixed between them (no pickup required) and must cooperatively move it from a random initial position in Room A to a fixed goal location within the same room (Fig. 1).
2. Task 2: Intra-Room Object Placement — The robots must cooperatively pickup and move the rod from a random starting point in Room A to a fixed goal location within the same room (Fig. 1).
3. Task 3: Inter-Room Transport — The robots must pick up and transport the same rod from a random position in Room A to a fixed goal position in Room B by navigating it through a narrow doorway that is shorter than the rod's length (Fig. 2).

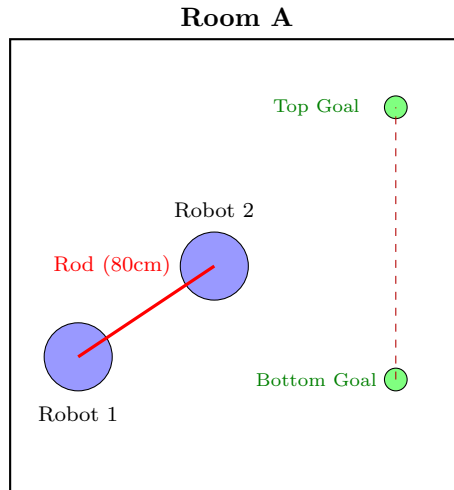


Figure 1: Task1 and Task2 top view. Green circles indicate the target positions for the rod ends.

Scenario Description

1. Environment Layout:
 - Room A and Room B are square rooms of size $2\text{m} \times 2\text{m}$.
 - The rooms are connected by a central doorway that is 50cm wide and centered on the shared wall.
 - The rod is a rigid object with length of 80cm.
2. Robots:
 - Two mobile robots, each of cylindrical shape with radius of 20cm and height of 10cm.

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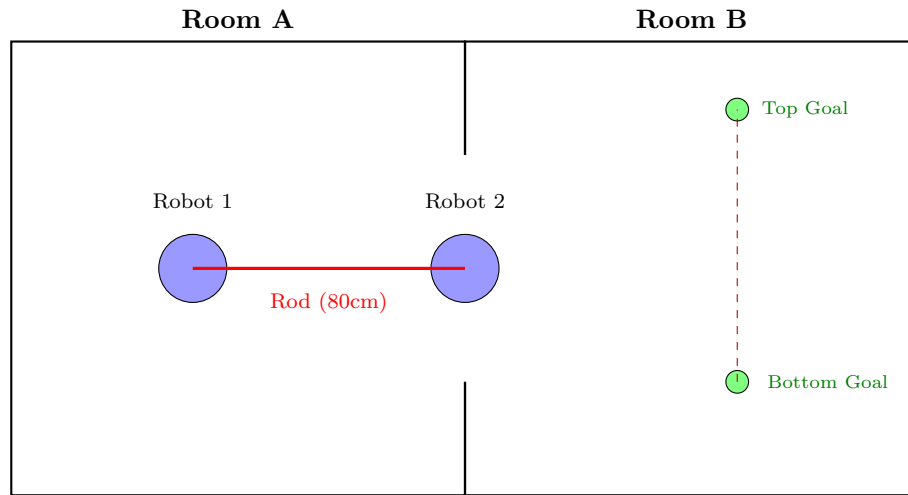


Figure 2: Task3 top view. Green circles indicate the target positions for the rod ends in Room B.

- Robots are capable of differential drive movement and independent wheel control on either side.
- A top-down vision-based system provides real-time information about the positions of the robots and the rod. This system communicates location data to the robots. Robots then use this location data to navigate the environment, avoid collisions with each other and with the walls, and locate the rod and goal positions.
- A robot successfully "picks up" the rod if the center of the robot reaches the corresponding end of the rod and remains there for a short time.
- Once both robots are correctly positioned at opposite ends, the rod becomes rigidly fixed between them and cannot be detached during transport.

3. Object Constraints:

- It must not touch or collide with the walls or the doorway frame.

Implementation Guidelines and Expected Delivery

Here is a breakdown of implementation steps for each task:

Core Setup

• Simulation Environment:

- Implement the basic environment using a suitable framework (e.g., Gymnasium, PyBullet).
- Create Room A ($2\text{m} \times 2\text{m}$).

• Robot Modeling:

- Implement two cylindrical mobile robots (radius 20cm).

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- Implement differential drive movement.
- Simulate the top-down vision system input (providing robot and rod positions).
- **Object Modeling:**
 - Implement the rigid rod object (length 80cm).
- **Basic Physics & Collision:**
 - Implement collision detection between robots, the rod, and walls.

Task 1: Coordination with Fixed Attachment

- **Goal Definition:** Define the fixed goal location in Room A.
- **Fixed Attachment:** Implement the logic where the rod starts fixed between the robots.
- **RL Algorithm (Task 1):**
 - Develop the core RL algorithm from scratch.
 - Define state representation (robot positions, rod position/orientation, goal location).
 - Define action space (robot movements).
 - Design a reward function encouraging cooperative movement towards the goal while avoiding collisions.
- **Training & Evaluation:** Train the agent for Task 1 and evaluate its ability to move the rod to the goal.

Task 2: Intra-Room Object Placement

- **Pickup Mechanism:**
 - Implement the "pickup" logic: robots moving to the rod ends and staying for a short duration.
 - Implement the rigid attachment logic once both robots pick up the rod.
- **RL Algorithm (Task 2):**
 - Adapt the RL algorithm (or develop a new phase) to handle the initial pickup phase, followed by transport.
 - Modify the state representation and reward function to include the pickup sequence.
- **Training & Evaluation:** Train the agent for Task 2 (pickup and transport) and evaluate performance.

Task 3: Inter-Room Transport

- **Environment Expansion:**
 - Add Room B ($2\text{m} \times 2\text{m}$) to the simulation.
 - Implement the 50cm wide doorway connecting the rooms.
- **Goal Definition:** Define the fixed goal location in Room B.

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- **RL Algorithm (Task 3):**
 - Modify the RL algorithm to handle the complex navigation challenge.
 - The reward function must penalize collisions with the doorway frame.
 - The agent needs to learn coordinated movements to pivot the rod through the narrow opening.
- **Training & Evaluation:** Train the agent for Task 3 and evaluate its ability to successfully transport the rod from Room A to Room B.

Notes

- You are supposed to implement the simulation environment and free using frameworks such as Gymnasium, PyBullet, MuJoCo, etc.
- You are free to use any reinforcement learning algorithm as long as you develop it from scratch and implement all core components yourself and avoid pre-built RL agents and libraries.
- Prepare a final report detailing the methodology, implementation, results, and analysis for each phase/task.